

October 5, 2026

The Editor-in-Chief
Computer Methods and Programs in Biomedicine

Dear Editor,

I am submitting the manuscript “Recognising the phone, not the throat: deconfounded evaluation and shortcut mitigation for smartphone pharyngitis classification” for consideration as a full-length article in *Computer Methods and Programs in Biomedicine*.

Multi-site medical image datasets often link the acquisition site to the outcome, and models can then score well by recognising the site. PGUPharyngitis, released in *Scientific Data* in 2025, is the largest public dataset of smartphone throat photographs (742 patients, two cities, one phone model per city), and bacterial prevalence differs threefold between its two phones. I show that the phone alone predicts the physicians’ label with an AUC of 0.67, above the published deep-learning baseline, and that image features identify the phone with an AUC of 0.99.

The manuscript then makes two methodological contributions that apply to any multi-site imaging dataset:

- *A validated deconfounded AUC.* Weighting each patient by $P(y)/P(y \mid \text{site})$ makes the label independent of the site, so any site-only predictor scores exactly 0.5 while comparisons across sites are kept. In a simulation with a known answer it was unbiased in every setting, and on the dataset resampled to different prevalence gaps it stayed within 0.02 of the shortcut-free AUC while pooled AUC rose from 0.54 to 0.78. Under this metric, image, symptom and fusion models all fall from about 0.70 to 0.52–0.58, and the 0.11 AUC advantage of images over symptoms disappears. We compare it with the covariate-adjusted AUC used in epidemiology and explain when the two differ.
- *A controlled comparison of shortcut mitigation.* Thirteen variants of six method families (colour constancy, per-site standardisation, linear concept erasure, reweighting, covariate adjustment and adversarial training) were tested on two backbones under one nested cross-validation protocol, together with end-to-end GroupDRO and adversarial fine-tuning, each with a probe of the site information left behind. Methods that remove the site lower the pooled AUC to the deconfounded level but none raises the deconfounded AUC, and adversarial training lowers AUC while leaving the site 97–98% identifiable, which would be mistaken for successful debiasing without a probe.

The study also provides the first multimodal (image plus symptom) benchmark on this dataset. All code, fold assignments and out-of-fold predictions are public, and the metrics and probes are released as a small Python package that computes them for any multi-site study from its predictions alone. The study follows TRIPOD+AI, and the completed checklist is included.

I believe the work fits the journal’s focus on computational methods for biomedical data, because it gives readers a simple evaluation tool and evidence on which debiasing methods do and do not work when site and outcome are confounded.

I confirm that this manuscript is original, has not been published, and is not under consideration elsewhere. I am the sole author. The study received no specific funding and I have no competing interests. It uses only the anonymised public dataset, whose collection was approved by a research ethics committee with informed consent. The use of AI-assisted tools is declared in the manuscript.

Thank you for considering this submission.

Yours sincerely,

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